

Unit - 1

Fintech – Financial Markets & Services

Fintech for Financial Mkts

Financial markets is a platform where buyers and sellers engage in the sale and purchase of financial products such as securities, commodities, and currencies. Stocks, bonds, and their derivatives are considered to be securities whereas commodities may include such items as valuable metals and agricultural goods, just but to mention a few.

One of the roles of financial markets is to set prices for the trade of securities, commodities, and currencies globally. Additionally, they help companies raise funds required for their smooth running and in order meet their financial obligations.

People trade financial securities, commodities, and value at low transaction costs and at prices that reflect supply and demand in financial markets. Securities include stocks and bonds, and commodities include precious metals or agricultural products.

Financial markets are broken down into various components based on the asset that is traded and the length of financing offered:

- **Capital Markets:** Capital markets are concerned with raising capital for entities through issuing equity stock or long-term debt. Let us explore some of the elements of capital markets.

a) **Stock markets:** They are markets that companies leverage to raise capital by creating shares. The shares are sold to external investors, who can either be corporations or individual buyers. Companies can sell the shares on their own or engage the services of stockbrokers.

b) **Bond markets:** When organizations need large-scale financing, they may resort to bonds. Bonds are borrowing instruments which generate a fixed interest rate after a predefined period. Different types of bonds include treasury bonds, corporate bonds, and municipal bonds.

- **Commodity Markets:** This is where companies trade in natural resources such as oil, corn or gold, for short-term financing
- **Money Markets:** Money market instruments are also referred to us 'paper'. They entail trading of financial instruments which are highly liquid and have short-term maturity dates. The maturity dates of papers could be as short as a few hours but not more than one year.
- **Derivatives Markets:** These are markets that facilitate the trading of financial instruments like futures contracts or options which are derived from other forms of assets like stocks, bonds, commodities, and market index price
- **Forward Markets:** A forward market is one where participants agree bilaterally, to buy or sell an asset at a future date at an agreed-upon price. The trades are executed over-the-counter (OTC). The contract features are bilaterally/mutually agreed upon rather than being standardized contracts issued by the exchange.

- **Insurance Markets:** Insurance markets are a means of protection against financial loss or any other uncertain loss. Insurance companies compensate to the insured party in case of occurrence of the insured loss.
- **Foreign Exchange Markets:** Forex markets deal with the buying and selling of foreign currencies. Institutions that have exposure to foreign currencies, due to their import and export transactions, trade in forex markets to hedge their exposure to protect themselves from potential losses.
- **Mortgage Markets:** They are classified as primary and secondary mortgage markets. In a primary mortgage market, mortgage brokers, commercial banks, and credit unions give out mortgage loans needed to acquire property. A secondary mortgage market involves the buying and selling of existing home loans, which have been bundled together and traded as mortgage-backed securities.

Key segments of Financial Mkts

- **Buy-side:** The buyers are the demand-creators of financial markets. Buyers can be private investors, mutual and pension funds, corporate and insurance companies, among others. Buyers actively participate in predicting the prices of securities and commodities.

Speculative or hedge buyers buy with the hope that the value of the commodity will go up so that they can later sell it at a higher price and make profits. The mutual funds are a pool of money that has been funded by owners or shareholders of different holdings and managed by professionals.

The pension funds, receive money from employees and individuals with the aim of covering for their retirement. Unlike mutual companies, pension funds are accessible upon the retirement of an individual.

Managers of both mutual funds and pension funds buy bonds and shares with the intention of generating interest for their investors. Other buyers include shareholders buying into a share issue and foreign currency bureaus.

- **Sell-side:** The sell-side creates the supply function of the market. They place and promote securities and commodities in the financial markets. Sellers include individuals, corporate and representatives, also known as agents.

Some investment banks provide their customers; mostly account owners, information about available securities and also issuer stocks and bonds on behalf of the selling companies.

A *stockbroker is a middleman* in the financial markets. The stockbroker can either be a company or an individual who purchases or buys stock and securities on behalf of another person. The advantage of engaging a stockbroker is that he is fully conversant with the news in the market can make the right decision at the right time and with a high degree of accuracy.

Stockbrokers represent both the individual and retail brokers, and their reward is the commission, paid by both the buyers and the sellers. Companies are also essential selling

players. Blue-chip companies issue shares to the public as a way of funding the company. Other securities sold by companies include debentures which yield interest to the buyers and rights shares.

- **Market infrastructure:** In order for the financial money markets to perform, they need a platform. Various institutions act as the platform and are therefore regarded as market infrastructure. Examples of such institutions are the Capital Market Authorities which hosts both the buyers and sellers of the markets. The institutions are also used by regulators to streamline the activities of the market makers.

Intermediary banks also play a critical role in providing infrastructure, especially in trading on the foreign currencies.

- **Regulators:** Regulators are tasked with the duty of making sure that the market is balanced. In most countries, regulators are the Central bank. The regulator for the US Government, for instance, is the Federal Reserve System. Regulators also control the supply of a country's currency and the interest rates to avoid inflation.

Financial Mkt Activities

- **Primary and new issue:** Primary issue means offering to the public securities that have not previously traded on the market. These can be in form of new stock, bond or debentures. An example of a primary issue is when privately owned companies goes public and introduce their shares to the market.
- **Trade:** Trade happens when a willing buyer meets a willing seller. In the financial markets, this mostly occurs in the broker's office or in the national exchange of that particular country. The price is determined by the demand and supply law. The economic and political situations of a country also play an essential role in determining the prices of securities. A rise of interest rates, for instance, can cause a demand rise of bonds issued then.
- **Clearing:** This entails all the activities that take place from the time a payment commitment is made to the point it is approved. The regulatory bodies are responsible for making sure that the instruments used are authentic and duly authorized by the right persons.
- **Settlement:** When the buyer and the seller agree on the price of the commodity, a contractual obligation is formed. To complete the contract, the seller exchanges the product with the buyer for the agreed value. This is referred to as settlement. When dealing with securities, settlement is fast because most of the time is digitalized.
- **Investment research:** Before investing in a security, an investor needs as much information as possible about the investment. Audit organizations carry out such audits and provide the report to the interested parties. Such statements help the investors to choose whether to invest and also determine the optimal investment portfolio.

- **Investment management:** Most of the times, investors lack the knowledge or the time to manage their interests. They, therefore, engage proficient administrators who manage investments either as stand-alone or portfolios with a promise to generate benefits. Real estate managers are a good example of investment managers.
- **Securities Pledging:** Securities act as collaterals for guaranteeing repayment of loans. A pledge is created in favour of the lender to facilitate disbursement of loan. Once the loan is repaid, the lender releases the pledge and the securities are free and revert to the borrower.
- **Securities lending:** Securities lending imparts liquidity to the market. It generates returns to the holders who do not intend to sell the securities. Nevertheless, they lend their securities to other market players and get the same after a certain period. They collect collaterals of higher value for the securities lent. They get interest/compensation for the period lent.

Capital Market:

A capital market is an important segment of the broader financial system where investors, institutions (like banks, corporates, investment firms) and governments can raise long-term funds by participating in the issuance and buying and selling of financial securities. These securities typically include stocks and bonds.

Functions of Capital Markets

- **Links investors and borrowers:** The role of capital markets is to act as a middleman between those who have investible surplus and those who require money for their projects.
- **Capital Formation:** An essential component of the capital market is capital formation. It provides funds on time to meet the financial needs of various economic sectors.
- **Mobilizing Savings** for investment is a key function of capital markets. Capital markets act as a conduit for channelling personal or institutional savings into productive investments. Here, individual and institutional investors purchase securities like stocks, bonds, or mutual funds, effectively converting their savings into investment capital.
- **Control security prices:** Capital markets promote the stability and systematic pricing of securities. It assures fair and transparent dissemination of prices to facilitate trading. The borrower is charged a standard interest rate or a minimum interest rate. Due to this, security prices in the economy stabilise.
- **Risk Diversification** in capital markets is a strategy to spread investment across various financial instruments, sectors, or markets to minimize the risk of loss. By investing in a mix of assets like stocks, bonds, and mutual funds, investors can reduce the impact of potential losses in any one area. This is based on the principle that not all markets or sectors move in the same direction simultaneously. Diversifying

investments helps balance the portfolio, ensuring that a downturn in one investment is offset by stability or gains in others.

- **Offers opportunities to investors:** The capital markets provide a variety of financial instruments to meet the demands of each investor. Although there are some risks, investors can obtain a better rate of return by investing in the capital market.
- **Reduces transaction time and cost:** In the capital market, the entire trading procedure is expedited, streamlined, and costs less. Every step of the trading process is automated, which speeds up the entire procedure.
- **Capital liquidity:** On the capital markets, investors can sell their assets to a third party for cash.
- **Information Dissemination** is a key function of capital markets, ensuring transparency and informed decision-making. These markets act as hubs for the distribution of vital financial information, including company earnings, stock performance, market trends, and economic indicators. This flow of information enables investors to analyze potential investments more accurately and make educated choices. It also ensures that all market participants have equal access to critical data, helping maintain a level playing field and promoting overall market efficiency and investor confidence.
- **Facilitating economic growth** is a fundamental capital market function. Capital markets channel savings and investments into productive uses, such as funding new business ventures, supporting infrastructure projects, or expanding existing companies. This influx of capital stimulates business activities, creates jobs, and drives technological advancements. By efficiently allocating resources where they can yield the highest returns, capital markets play a pivotal role in stimulating the overall economic development and growth of a country, contributing to a healthier, more robust economy.
- **Wealth Generation** is another function of capital markets where individuals and institutions grow their wealth by investing in various financial instruments. Capital markets offer a range of investment options like stocks, bonds, and mutual funds, which have the potential for capital appreciation and dividend income. Over time, these investments can yield significant returns, contributing to wealth accumulation for investors. This function benefits individual investors and fuels economic growth by reinvesting and circulating wealth within the economy.
- **Resource Allocation in capital markets** refers to the efficient distribution of financial resources among various entities and projects. Capital markets channel funds from savers and investors to businesses, governments, and other institutions that require capital for growth, development, or operational purposes. By aligning capital with the most promising and productive opportunities, these markets ensure that funds are invested where they can yield the highest returns. This process is crucial for optimal economic growth, as it directs capital to where it's most needed and can be most effectively utilized.

- The functions of the capital market are vital for economic stability and growth. They enable the efficient allocation of resources, provide long-term funding for businesses and governments, and offer investment opportunities for individuals and institutions. Capital markets contribute to financial stability and foster economic development by ensuring liquidity and facilitating securities trading.

Types of Securities traded in Capital Markets

Security: A Security is a financial instrument or a financial asset of value, that can be traded in the open market, e.g., a stock, bond, options contract, or units of a mutual fund etc.

Broadly, there are 4 types of securities, namely:

1. **Equity securities:** Equity securities represent ownership interest held by shareholders in a company. In other words, it is an investment in an organization's equity stock to become a shareholder of the organization.

The difference between holders of equity securities and holders of debt securities is that the equity holders are not entitled to a regular payment, but they can profit from **capital gains by selling** the stocks. Another difference is that equity securities provide **ownership rights to the holder** so that he becomes one of the owners of the company, owning a stake proportionate to the number of acquired shares.

In the event a business faces bankruptcy, the equity holders can only share the residual interest that remains after all obligations have been paid out to debt security holders. Companies regularly distribute dividends to shareholders sharing the earned profits coming from the core business operations, whereas it is not the case for the debtholders.

2. **Debt securities:** Debt securities, or fixed-income securities, represent money that is borrowed and must be repaid with terms outlining the amount of the borrowed funds, interest rate, and maturity date. In other words, debt securities are debt instruments, such as bonds (e.g., a government or a corporate bond) or a Certificate of Deposit that can be traded between parties.

Debt securities, such as bonds and certificates of deposit, as a rule, require the issuer/borrower to make the regular interest payments, as well as repayment of the principal amount alongside any other stipulated contractual rights. Such securities are usually issued for a fixed term, and, in the end, the issuer redeems them.

A debt security's interest rate on a debt security is determined based on a borrower's credit history, track record, and solvency – the ability to repay the loan in the future. The higher the risk of the borrower's default on the loan, the higher the interest rate a lender would require to compensate for the amount of risk taken.

3. **Hybrid securities:** Hybrid security, as the name suggests, is a type of security that combines characteristics of both debt and equity securities. Many banks and organizations turn to hybrid securities to borrow money from investors.

Similar to bonds, they typically promise to pay a higher interest at a fixed or floating rate until a certain time in the future. Unlike a bond, the number and timing of interest payments are not guaranteed. They can even be converted into shares, or an investment can be terminated at any time.

Examples of hybrid securities are preferred stocks that enable the holder to receive dividends prior to the holders of common stock, convertible bonds that can be converted into a known amount of equity stocks during the life of the bond or at maturity date, depending on the terms of the contract, etc.

4. **Derivatives:** Derivative securities are financial instruments whose value is derived from an underlying security. The underlying can be assets, such as stocks, bonds, currencies, interest rates, market indices, and goods. The main purpose of using derivatives is to hedge to minimize risk.

Role of Capital Mkts

Mobilizing Savings: Capital markets help in aggregating savings from different sectors of the economy and make them available for investment in productive activity. The mobilization ensures the creation of capital that drives economic activities and growth.

Facilitating Investment: They offer investors a platform to spread their portfolios by investing in any kind of security. They facilitate the use of resources by the more productive sectors, and hence, there is economic development.

Economic Stability: This stability of the financial system results from the possibility of long-term investment. The capital markets reduce economic volatility as they help allocate resources efficiently.

Job Generation: The Indian capital market influences employment directly through capital markets' growth. As capital markets develop businesses and industries, there will be greater opportunities for job creation.

Capital Market and Innovation/Empowerment: Availability of capital stimulates innovation and entrepreneurship. Capital markets finance new initiatives, such as innovative projects or new ventures. Such financing contributes to technological changes and economic development.

Key Drivers for Growth of Capital Markets in India

Economic Reforms: The liberalization policies, reforms in regulation, and new financial instruments introduced have all added value to the **capital market growth in India**.

Technological Advancements: Technology integration in trading and settlement processes has made capital markets more accessible and efficient. Online trading platforms have democratized market participation.

Foreign Investments: FDI and FII have played a crucial role in the growth of capital markets. The inflow of foreign capital has enhanced liquidity and market depth.

Government initiatives: Its policies for easy doing business, promotion of startups and encouraging investments have further contributed to the growth of **capital markets in India**.

Capital Markets in Economic Growth

Capital markets play a multilateral role in economic development. It supports economic growth in various ways through facilitating capital flows.

Infrastructure Development: Major infrastructure projects such as roads and highways require infusions of capital. These forms of development remain important for growing the economy in that they lower logistics costs and improve connectivity, which increases productivity.

Corporate Growth: Access to capital allows companies the opportunity to engage in expansion efforts, enter markets, and innovate through research. Corporate growth culminates to higher economic activities and employment.

Wealth Creation: Investment in capital markets opens avenues for wealth creation for individual investors. The wealth created increases consumption and demand, which is the driving force for economic growth.

Government Financing: The government may utilise capital markets in raising money on bond issues, which then shall be spent directly on development undertakings as well as various social welfare activities and public programs. Therefore, the fund sources would aid economic development.

MONEY MARKETS

The money market is a financial market wherein short-term financial assets are traded between institutions and traders. The market offers very high liquidity as the assets can be easily be converted into cash. Thus, it helps businesses and the government in meeting their working capital requirements.

Investments here fall under the range of a day to a year, with the risks and gains both being low.

- Money Market refers to the **financial segment for the trade of liquid and short-term financial assets** that can be easily converted into cash. Businesses and governments particularly benefit from this market as it helps in meeting their working capital requirements.
- Unlike a capital market where long-term securities are exchanged, here the trades investments range from a day, three months, 365 days, etc.

- Longer the period of investment, higher the risk and potential gains. As such, the assets traded here come with low risk and gains.
- Due to high transaction value, market investors are largely insurance companies, governments, NBFCs, banks, credit institutions, etc. Retail investors invest using money market funds and accounts.
- Popular investment instruments include treasury bills, Certificate of deposit, commercial papers, Bills of Exchange.

Why it is called Money Mkt: The money market deals in highly liquid, very safe, short-term debt securities and these attributes make them virtual cash equivalents. They can be exchanged for cash at short notice.

Importance of Money Mkts: The money market keeps the financial economy running smoothly. It allows savers to lend money to those in need of short-term loans and it allocates capital toward its most productive use.

These loans are often made overnight or for a matter of days or weeks. They're needed by governments, corporations, and banks to meet their near-term obligations or regulatory requirements. And they allow those with some excess cash on hand to earn a small amount of interest.

Instruments traded in Money Markets:

Treasury bills are issued by government at a discount to the face value of the bond. The difference between the Face Value and the Issue price is the interest earned by the holder

Certificate of Deposit (CD) are issued by commercial banks at a discount to the face value. The difference between the Face Value and the Issue price is the interest earned by the holder

Commercial Papers are issued by Corporates at a discount to the face value. The difference between the Face Value and the Issue price is the interest earned by the holder

Call Money: Overnight borrowings by commercial banks: Call Money refers to **inter-bank borrowing and lending for a very short period**, typically overnight to upto 14 days. The Call Money or Money at Call enables banks and financial institutions to manage their short-term liquidity requirements. The rate at which money is borrowed in these markets is called the **Call Money Rate**. The Call Money Rate **keeps changing on an hourly basis**, depending on the demand and supply.

Repurchase Agreements: Repos are agreements between banks and Reserve Bank

Banker's Acceptances (BA): Banker's acceptance is a financial instrument in which a bank guarantees payment to a third party at a future date, rather than the account holder guaranteeing the payment. The bank assumes responsibility for paying the third party if the account holder defaults on the payment. Banker's acceptance is typically used in cross-border trade to protect exporters from counterparty default risk. The bank assesses the account holder's credibility on various grounds, including their credit history. If the account holder meets the bank's criteria, the bank will assume responsibility on their behalf.

Money market participants

- Central and State Govts.
- Commercial banks
- Non Banking Finance Companies (NBFCs)
- PSUs & Corporates
- Provident Fund Trusts & Pension Funds
- Institutional Investors like Mutual funds

Demand for Money market instruments is largely driven by

- **Statutory Liquidity Ratio (SLR) Requirements** as stipulated by regulatory agencies leading to Banks and NBFCs being active participants. SLR investments are required to meet any financial emergency, as the investments are highly liquid and can be quickly converted to cash.
- Provident Funds, Pension Funds and Gratuity funds too have to invest a part of their corpus in money market instruments to meet any repayment obligations
- Corporates and mutual funds invest in MM frequently to park their surplus cash balances to generate returns
- Banks and Financial Institutions invest/borrow in money markets for periods as low as 1 day to 1 week to tide over short term financing requirements and/or comply with regulatory regulations like reporting Friday.

Trading platform is managed by Clearing Corp of India Ltd. (CCIL - a subsidiary of RBI) on which funds are borrowed/lent based on the bids and offers made by the members on the platform. The Telephone driven market is still thriving with deals struck on bilateral basis. However, all the deals have to be reported to RBI through CCIL platform

Functions of Money Markets

Short-Term Borrowing and Lending

The money market is a platform for institutions to meet their short-term funding needs. Banks, corporations, and governments borrow money for a brief period, usually less than a year. This bridges gaps in their immediate cash flows requirements. On the other hand, lenders provide these funds in exchange for interest income. It allows them to earn a return on their surplus funds.

Price Discovery

Instruments such as treasury bills, CDs and commercial paper, are actively traded. The issue prices of these instruments reflect prevailing market interest rates. This price discovery mechanism aids in determining the cost of short-term borrowing and the potential returns for lenders. As a result, the money market plays a crucial role in assessing the overall health of the financial system and the broader economy.

Liquidity Management

Investors seeking secure and easily accessible avenues for excess funds turn to the money market. Money market instruments are highly liquid, meaning they can be quickly converted into cash without significant loss in value. This liquidity feature ensures that investors can swiftly access their funds in need. It makes the money market a reliable option for managing short-term liquidity requirements.

Financing Trade and Commerce

International trade is supported by providing financing solutions for exporters and importers. Instruments like letters of credit and banker's acceptances facilitate cross-border transactions. Importers can use these instruments to assure payment to exporters, while exporters can obtain funds before goods are delivered. This function promotes smoother trade activities and reduces the risk of international transactions.

Similarly, **domestic trade** is also supported by the financial solutions in the form of Letters of Credit, Banker Guarantee and Bills of Exchange wherein the supplier can secure his payment the moment goods are received by the buyer. However, the buyer can secure credit from his bank for a certain period within which he can sell the product and realise payments to pay the bank.

Reserve Management for Banks

Banks must maintain certain reserves to ensure their stability and compliance with regulations. The money market allows them to earn interest on these idle funds. Banks can optimise their reserves by investing in money market instruments and generate a modest return while fulfilling their regulatory obligations.

Yield Enhancement

Investors seeking conservative yet yielding investment options find money market instruments attractive. These instruments provide moderate returns compared to traditional savings accounts or certificates of deposit. It makes them a preferred choice for individuals and organizations looking to enhance their yields without taking excessive risk.

Central Bank Operations

[Central banks](#) utilize the money market to implement [monetary policy](#) measures. They buy or sell money market instruments through [open market operations](#) to influence the money

supply and interest rates. By adjusting these key factors, central banks aim to achieve economic goals such as controlling inflation, stimulating economic growth, or maintaining price stability.

Risk Mitigation

Corporations and financial institutions manage their interest rate and liquidity risks through the money market. Derivatives and short-term financing options, such as repurchase agreements (repos), enable them to hedge against adverse market movements. These risk management tools enhance financial stability and enable businesses to navigate uncertain market conditions more effectively.

Support for Government Financing

Governments raise funds for short-term budgetary needs by issuing Treasury bills in the money market. These bills serve as a means for governments to cover temporary deficits, manage cash flow fluctuations, and meet immediate financial obligations. The money market provides a reliable source of funds for governments while offering investors a secure investment option.

Investor Diversification

The money market broadens investment opportunities for individuals and institutional investors. Investors can diversify their portfolios beyond traditional [stocks](#) and bonds by participating in money market activities. This diversification reduces overall portfolio risk and provides a balanced investment approach that aligns with varying risk tolerances and financial goals.

Importance of Money Market

A developed money market plays an important role in the financial system of a country by supplying short-term funds adequately and quickly to trade and industry. The money market is an integral part of a country's economy. Therefore, a developed money market is highly indispensable for the rapid development of economy. A developed money market helps the smooth functioning of the financial system in any economy in the following ways:

- **Development Of Trade And Industry:** Money market is an important source of financing trade and industry. The money market, through discounting operations and commercial papers, finances the short-term working capital requirements of trade and industry and facilitates the development of industry and trade both — national and international.
- **Development Of Capital Market:** The short-term rates of interest and the conditions that prevail in the money market influence the long-term interest as well as the resource mobilization in capital market. Hence, the development of capital depends upon the existence of a development of capital money market.
- **Smooth Functioning of Commercial Banks:** The money market provides the commercial banks with facilities for temporarily employing their surplus funds in easily

realizable assets. The banks can get back the funds quickly, in times of need, by resorting to the money market. The commercial banks gain immensely by economizing on their cash balances in hand and at the same time meeting the demand for large withdrawal of their depositors. It also enables commercial banks to meet their statutory requirements of cash reserve ratio (CRR) and Statutory Liquidity Ratio (SLR) by utilizing the money market mechanism.

- **Effective Central Bank Control:** A developed money market helps the effective functioning of a central bank. It facilitates effective implementation of the monetary policy of a central bank. The central bank, through the money market, pumps new money into the economy in slump and siphons it off in boom. The central bank, thus, regulates the flow of money so as to promote economic growth with stability.
- **Formulation Of Suitable Monetary Policy:** Conditions prevailing in a money market serve as a true indicator of the monetary state of an economy. Hence, it serves as a guide to the Government in formulating and revising the monetary policy then and there depending upon the monetary conditions prevailing in the market.
- **Non-Inflationary Source Of Finance To Government:** A developed money market helps the Government to raise short-term funds through the treasury bills floated in the market. In the absence of a developed money market, the Government would be forced to print and issue more money or borrow from the central bank. Both ways would lead to an increase in prices and the consequent inflationary trend in the economy.

Unit 2

Risk & Return

- **Background:** The result of increasingly global markets is that risk may originate with events thousands of miles away that have nothing to do with the domestic market. Information is available instantaneously, which means that change, and subsequent market reactions, occur very quickly
- it is important to ensure financial risks are identified and managed appropriately. Preparation is a key component of risk management
- **Risk** is the outcome of an action taken or not taken, in a particular situation which may result in loss or gain. It is termed as a chance of loss or, exposure to danger, arising out of internal or external factors that can be minimised through preventive measures.
- **Risk** is the likelihood of losses resulting from events such as changes in market prices. Events with a low probability of occurring, but that may result in a high loss, are particularly troublesome because they are often not anticipated. Put another way, risk is the probable variability of returns.
- **Uncertainty:** absence of certainty or something which is not known. It refers to a situation where there are multiple alternatives resulting in a specific outcome, but the probability of the outcome is not certain. This is because of insufficient information or knowledge about the present condition, and making it hard to define or predict the future outcome or events.
- **Uncertainty** cannot be measured in quantitative terms through past models. Therefore, probabilities cannot be applied to the potential outcomes, because the probabilities are unknown

Differences between Risk and Uncertainty:

- Risk is defined as the situation of winning or losing something worthy. Uncertainty is a condition where there is no knowledge about the future events.
- Risk can be measured and quantified, through theoretical models. Conversely, it is not possible to measure uncertainty in quantitative terms, as the future events are unpredictable
- The potential outcomes are known in risk, whereas in the case of uncertainty, the outcomes are unknown
- Risk can be controlled if proper measures are taken to control it. Whereas, uncertainty is beyond the control of the person or enterprise, as the future is uncertain
- Risk Minimization can be done, by taking necessary precautions as opposed to uncertainty that cannot be minimised.
- In risk, probabilities are assigned to a set of circumstances which is not possible in case of uncertainty

Conclusion:

- An old saying, “**No Risk, No Gain**”, for an enterprise to survive in the long run, calculated risks are inevitable where the probability of loss is comparatively less, and the chances of gains are higher.
- **Uncertainty** is inherent in every business which cannot be avoided, and the business person has no idea about what will happen next, i.e. the outcome is unknown

Main sources of Financial Risk

- Financial risks arising from an organization's exposure to changes in market prices, such as interest rates, exchange rates, and commodity prices
- Financial risks arising from the actions of, and transactions with, other organizations such as vendors, customers, and counterparties in derivatives transactions.
- Financial risks resulting from internal actions or failures of the organization, particularly people, processes, and systems.

Categories of Risk:

1. Financial Risk
2. Operational Risk
3. Strategies Risk
4. Compliance Risk
5. Reputation Risk
6. Environmental Risk
7. Health & Safety Risk
8. Technology Risk
9. Political Risk
10. Social Risk

Financial Risks

1. **Market Risk:** It is a fall in investment value due to market risk factors, that include equity risk, interest rate risk, currency risk and commodity price risk.
2. **Credit Risk:** Also known as Default risk. It is the risk arising out of the credit/loan given to an individual or a company, who may be unable to repay the interest or principal or both, on its debt obligations. Govt. bonds have lowest credit risk, while lowly rated corporate bonds carry high credit risk.

Liquidity Risk: It is the inability of an individual or a company to meet its financial obligations due to lack of funds. If an organization has insufficient liquidity to maintain its day-to-day

operations. While revenues and sales may be sufficient for long-term growth, if short-term cash is insufficient, liquidity issues may require decisions that are detrimental to long-term growth

Risk Measurement: Measures of Risk

- **Standard Deviation**
- Co-efficient of variation
- R-Square
- **Alpha**
- **Beta**
- Sharpe Ratio

Standard Deviation: Standard deviation is a statistical measurement that looks at how far individual points in a dataset are dispersed from the [mean](#) of that set. If data points are further from the mean, there is a higher deviation within the data set. It is calculated as the square root of the variance.

Features of SD:

- SD measures the dispersion of a dataset relative to its mean.
- SD is calculated as the square root of the variance.
- SD, in finance, is often used as a measure of the relative riskiness of an asset.
- A volatile stock has a high standard deviation, while the deviation of a stable blue-chip stock is usually rather low.
- Standard deviation is also used by businesses to assess risk, manage business operations, and plan cash flows based on seasonal changes and volatility.

Applications:

- When applied to the annual rate of return of an investment, it can provide information on that investment's historical volatility. This means that it shows how much the price of that investment has fluctuated over time.
- The greater the standard deviation of securities, the greater the variance between each price and the mean, which shows a larger price range.
- Standard deviation can also be used to predict performance trends. In investing, for example, an index fund is designed to replicate a benchmark index. This means that the fund will have a low standard deviation from the value of the benchmark.
- Aggressive growth funds often have a high standard deviation from related indices. This is because their portfolio managers make aggressive bets to generate higher-than-

average returns. This higher standard deviation correlates with the level of risk investors can expect from that index.

- Standard deviation is one of the key fundamental risk measures that analysts, portfolio managers, and advisors use
- The formula for Standard Deviation (σ) is:

$$\sigma = \sqrt{[\sum (x_i - \mu)^2] / (n - 1)}$$

Where:-

σ = Standard Deviation

x_i = Individual data points

μ = Mean of the data set

n = Number of data points

Σ = Summation symbol (sum of the squared differences)

Computation of Standard Deviation:

1. Calculate the mean of all data points. The mean is calculated by adding all the data points and dividing them by the number of data points.
2. Calculate the variance for each data point. The variance for each data point is calculated by subtracting the mean from the value of the data point.
3. Square the variance of each data point (from Step 2).
4. Sum of squared variance values (from Step 3).
5. Divide the sum of squared variance values (from Step 4) by the number of data points in the data set less 1.
6. Take the square root of the quotient (from Step 5).

Properties of Standard Deviation

- One key property of SD is **additivity**: The entire set of data points is taken into consideration for analysis rather than a selected few.
- **Scale invariance**: Useful in comparing the variability of datasets with different measurement units
- SD has **data symmetry & non-negativity**: means SD is always positive and symmetrically distributed around mean. Symmetry implies that deviations above the mean are balanced by deviations below the mean resulting in a balance of entire data set.

Risk Mitigation / Risk Management:

Risk mitigation is the process of reducing or eliminating potential risks, by implementing strategies, to minimize their impact on an organization or project. Techniques:

- **Risk Avoidance:** Eliminate or avoid the risk by changing plans or circumstances
- **Risk Transfer:** Shift the risk to another party, such as through insurance or outsourcing
- **Risk Reduction:** Implement controls or measures to reduce the likelihood or impact of the risk.
- **Risk Acceptance:** Accept the risk and monitor it, but take no action to risk or mitigate it.

Risk Premium:

Risk premium refers to the additional return an investor expects to receive from an investment to compensate for the risk associated with holding that asset.

In essence, it represents the difference between the expected return on a risky asset and the return on a risk-free asset, such as government bonds.

Risk premium plays a crucial role in investment decisions, as it helps investors determine the appropriate level of compensation for the risks they are taking. It also helps in evaluating potential investment opportunities, asset allocation, and performance measurement.

The risk premium is directly related to the expected return and risk associated with an investment. Higher risk investments generally require a higher risk premium to attract investors, as they compensate for the increased potential for loss.

Types of Risk Premium:

Business Risk refers to the uncertainty of a company's future cash flows, its longevity and sustainability.

Financial Risk refers to a company's ability to manage the financing of its operations.

Credit Risk Premium: The credit risk premium is the additional return investors require for investing in bonds or other fixed income securities with credit risk, compared to risk-free government bonds. This premium compensates investors for the possibility of default by the bond issuer.

Liquidity Risk Premium: The liquidity risk premium compensates investors for the risk associated with holding less liquid assets that cannot be easily converted to cash without incurring significant costs. This premium is higher for investments that are more difficult to sell or trade in the market. uncertainty related to an investor's ability to exit an investment, both in terms of timeliness and cost.

Exchange Rate Risk: refers to the *potential for loss that arises from fluctuations in the exchange rate between two currencies*. This risk arises when an investor or business conducts transactions in a foreign currency or holds assets denominated in a foreign currency

Equity Risk Premium: Difference between expected return on equity and Risk Free Rate

Country Risk Premium: The country risk premium is the additional return investors expect for investing in assets from a specific country, due to the risks associated with that country's economic, political, and regulatory environment. This premium varies across countries and can change over time as conditions evolve.

Factors influencing Risk Premium:

Market Conditions and Economic Environment: Risk premiums are influenced by prevailing market conditions and the broader economic environment. During periods of market volatility or economic uncertainty, risk premiums may increase as investors demand higher compensation for taking on risk.

Investor Risk Tolerance: Risk premiums also reflect the collective risk tolerance of investors. When investors are more risk-averse, they may demand higher risk premiums for holding risky assets, and vice versa.

Time Horizon: The time horizon of an investment also impacts risk premiums. Longer-term investments generally require higher risk premiums to compensate for the increased uncertainty over time.

Asset Class and Investment Characteristics: Different asset classes and individual investments exhibit unique risk characteristics, which can influence their associated risk premiums.

Assessment of credit worthiness and Gradation:

- The creditworthiness of organizations and other entities is assessed by credit rating agencies (CRAs). The agencies examine a debtor's capacity to pay back the loan and provide a credit risk rating to that debtor. There are 7 agencies operating in India
- 1. CRISIL
- 2. ICRA
- 3. CARE
- 4. Brickwork Rating
- 5. India Rating & Research
- 6. SMERA
- 7. Infomeric Valuation & Rating Pvt Ltd

Credit Rating:

Rating Symbols and Definitions for Issuer Rating symbols should have CRA's first name as prefix

- **AAA** – Issuers with this rating are considered to have the highest degree of safety regarding timely servicing of debt obligations. Debt exposures to such issuers carry lowest credit risk.
- **AA** – Issuers with this rating are considered to have high degree of safety regarding timely servicing of debt obligations. Debt exposures to such issuers carry very low credit risk.
- **A** – Issuers with this rating are considered to have adequate degree of safety regarding timely servicing of debt obligations. Debt exposures to such issuers carry low credit risk
- **BBB** – Issuers with this rating are considered to have moderate degree of safety regarding timely servicing of debt obligations. Debt exposures to such issuers carry moderate credit risk.
- **BB** – Issuers with this rating are considered to have moderate risk of default regarding timely servicing of debt obligations.
- **B** – Issuers with this rating are considered to have high risk of default regarding timely servicing of debt obligations.
- **C** – Issuers with this rating are considered to have very high risk of default regarding timely servicing of debt obligations.
- **D** – Issuers with this rating are in default or are expected to be in default soon

Ex-Ante Returns

Refers to estimation or forecasting financial performance of a company based on historical data. Similarly, it may also involve forecasting interest rates based on historical data of variables impacting interest rates.

Ex Post Returns: Ex-post is the opposite of ex-ante. It is post facto. It takes into account the actual financial performance of a company.

A comparison is done between the estimation and actuals to gain insight on the variables that caused the difference.

Ex Ante Returns:

- Prediction before the events.
- Based on assumptions
- Future oriented
- Generally used to predict returns, business outcomes (mkt share) etc.
- Estimated Returns that can be expected (i.e. probability is involved).
- Useful in comparing estimates with actuals and learn lessons on variables that were at divergence vis-à-vis the actuals.

Computation of Ex-Ante Returns:

Business is uncertain in future. Hence, various scenarios are constructed. Probabilities are assigned to each scenario. For example:

State of Economy	Probability (P)	Returns (R)	P*R
Good	0.30	20% or 0.20	0.060
Average	0.50	15% or 0.15	0.075
Bad	0.20	-4% or -0.04	-0.008
		Mean Return $\bar{R}$	0.127 or 12.7%

Standard Deviation:

Probability	Returns	$R - \bar{R}$		Variance
0.3	20%	$0.20 - 0.127$	0.073	0.005329
0.5	15%	$0.15 - 0.127$	0.023	0.000529
0.2	-4%	$-0.04 - 0.127$	-0.087	0.007569
			Total	0.013427

Avg. Variance: $0.013427/2 = 0.0067135$

Std Deviation = Sqr Root of Variance = $\sqrt{0.0067135} = 0.0819$ or say 8.19%

Ex-Post Returns

- Refers to outcomes after the event
- Use this concept to project/forecast future.
- Ex-Post Returns calculated after taking starting and ending values of a period

Risk-Return Tradeoff:

- Refers to the relationship between Risk and Return.
- Linear Relationship between Risk & Return. i.e. If one variable $\uparrow$, the other variable also $\uparrow$

- An investment in FD of SBI would give lower returns but the risk is also lower. However, if the money is lent in unorganized market, the return would be higher but the risk will also be higher. For Example:

Particulars	Risk	Return
Fixed Deposit in SBI	↓ Low Risk	↓ Low Return
Fixed Deposit in Exide Batteries	↓ Moderately low risk	↓ moderately lower return
Equities	↑ Higher Risk	↑ Higher Return
Non Formal Lending	↑↑ Highest Risk	↑↑ Highest Return

- Generally, Risk is the independent variable and Return is the dependent variable.
- Risk & Return are the 2 sides of the same coin.

Influencing Factors:

- Risk Tolerance: Young (Higher) vs. old (Lower)
- Investment horizon: Building corpus (aggressive - equities) vs. Investing for Retirement (conservative – Debt).
- **Asset Allocation:** The Young and middle age is characterized by moderate to higher earnings with lesser responsibilities leading to higher risk appetite. During this phase, asset allocation would be more tilted towards Equities with higher return expectations. However, as middle age, mid life sets in, the earnings remain high but responsibilities are also higher leading to moderate risk appetite. During this phase, Debt allocation share is higher with moderate return expectations. As the end of productive work life sets in the risk appetite is at its lowest and the debt allocation is at its highest.
- **Portfolio diversification:** The investment portfolio has to be diversified so as to have the ability to absorb the shocks of any asset class in the portfolio.
- **Portfolio rebalancing:** Portfolio needs to be rebalanced periodically as and when the phase changes and tuned as per the risk appetite.

Evaluation of Risk-Return Tradeoff:

- **Alpha:** Refers to excess returns generated over the benchmark Index
- **Beta:** Refers to how an investment responds to changes in benchmark.

Unit – 3

Fintech - Overview

1. **Growth and Innovation:** Fintech has seen rapid growth due to technological advancements, digitization, and changing customer preferences. It has reshaped areas like payments, lending, insurance, and wealth management with innovative, customer-centric solutions.
2. **Market Size and Investment:** The fintech industry has attracted significant investments. For example, in India, the fintech market is expected to grow from USD 145.09 billion in 2025 to USD 550.21 billion by 2030. Globally, publicly traded fintechs had a market capitalization of \$550 billion as of July 2031.
3. **Key Players and Startups:** The fintech ecosystem includes a mix of established financial institutions and startups. In India, prominent fintech companies include Paytm, Pine Labs, PayU, and Faircent. These companies leverage technologies like AI, blockchain, and IoT to offer innovative financial solutions.
4. **Regulatory Support:** Governments and regulatory bodies have played a crucial role in supporting fintech growth. Initiatives like Digital India and favorable policies from the Reserve Bank of India have encouraged innovation and financial inclusion.
5. **Challenges and Future Trends:** Despite its growth, the fintech sector faces challenges such as regulatory hurdles, cybersecurity threats, and market volatility. However, the focus is shifting towards sustainable and profitable growth, with fintechs adapting to a more stable and steady pace.

Evolution of Fintech

The evolution of fintech can be traced back to the late 20th century, with significant milestones marking its journey:

1. **Early Beginnings (1970s-1980s):** The advent of computer systems and electronic banking laid the foundation for fintech. Innovations like ATMs and credit cards revolutionized banking by providing 24/7 access to funds and eliminating the need for cash transactions.
2. **Digital Transformation (1990s-2008):** The rise of the internet and e-commerce in the 1990s brought online banking and electronic payments to the forefront. The establishment of NASDAQ, the first digital stock exchange, and the introduction of SWIFT for cross-border payments were pivotal.
3. **Post-Financial Crisis (2008-2014):** The 2008 global financial crisis spurred the growth of fintech as consumers sought alternatives to traditional banking. This period saw the emergence of startups offering innovative solutions in payments, lending, and insurance.
4. **Modern Era (2014-Present):** The proliferation of smartphones and mobile apps has driven fintech's rapid expansion. Technologies like blockchain, artificial intelligence,

and big data analytics have enabled the development of digital wallets, peer-to-peer lending platforms, robo-advisors, and more.

Significance to Banking, Financial Services, and Insurance (BFSI) Sectors

Fintech has had a profound impact on the BFSI sectors, transforming the way financial services are delivered and consumed:

1. Banking:

- **Enhanced Customer Experience:** Fintech has introduced user-friendly mobile banking apps, digital wallets, and online banking platforms, making financial services more accessible and convenient.
- **Operational Efficiency:** Automation and AI-driven solutions have streamlined banking operations, reducing costs and improving efficiency.
- **Financial Inclusion:** Fintech has enabled underserved populations to access banking services through digital platforms, promoting financial inclusion.

2. Financial Services:

- **Innovative Solutions:** Fintech has revolutionized payments, lending, and investment management with innovations like peer-to-peer lending, crowdfunding, and robo-advisors.
- **Data-Driven Insights:** Big data analytics and AI have empowered financial institutions to offer personalized financial products and services based on customer behavior and preferences.
- **Risk Management:** Advanced technologies like blockchain and AI have enhanced risk management and fraud detection capabilities.

3. Insurance:

- **Digital Transformation:** Insurtech, a subset of fintech, has digitized insurance processes, from underwriting to claims processing, improving efficiency and customer experience.
- **Customized Products:** Data analytics and AI have enabled insurers to offer personalized insurance products tailored to individual needs.
- **Improved Accessibility:** Digital platforms have made insurance products more accessible to a broader audience, promoting financial security.

Fintech continues to drive innovation and reshape the BFSI sectors, offering new opportunities and challenges. Its significance lies in its ability to enhance efficiency, accessibility, and customer experience while fostering financial inclusion and innovation.

Fintech, or financial technology, is revolutionizing the way we manage and interact with our finances. Here are some key areas where fintech is being extensively used:

1. **Digital Payments and Mobile Wallets:** Platforms like Apple Pay, Google Wallet, and PayPal have transformed how we make transactions, allowing for quick and secure payments directly from our devices.
2. **Blockchain and Cryptocurrency:** Blockchain technology underpins cryptocurrencies like Bitcoin and Ethereum, enabling decentralized financial transactions without traditional intermediaries.
3. **Digital Lending and Credit:** Online lending platforms have made it easier for individuals and businesses to borrow money without the need for physical bank visits or extensive paperwork.
4. **Insurtech (Insurance Technology):** Fintech is also making waves in the insurance industry by streamlining processes, improving customer experiences, and offering personalized insurance products.
5. **Robo-Advisors and Wealth Management:** Automated platforms provide financial advice and investment management services, making it easier for individuals to manage their portfolios.
6. **RegTech (Regulatory Technology):** This area focuses on helping financial institutions comply with regulations more efficiently through the use of technology.

Fintech is not just limited to these areas; it continues to evolve and expand, bringing innovation and efficiency to various aspects of the financial industry.

Fintech is continuously evolving, and several emerging areas are poised to transform the financial landscape even further. Here are some of the most promising areas:

1. **Embedded Finance:** This involves integrating financial services into non-financial platforms, such as e-commerce or ride-sharing apps. For example, Shopify enables merchants to offer in-app payment options and loans through its Shopify Capital program.
2. **Decentralized Finance (DeFi):** DeFi is moving towards regulation, which will bring more stability and security to the sector. This will make it more attractive to institutional investors.
3. **Carbon-Conscious Financial Products:** Fintech is addressing climate concerns by offering tools that help consumers track and offset their carbon footprint. Mastercard's Carbon Calculator is one such example.
4. **AI-Driven Hyper-Personalization:** AI is being used to provide highly personalized financial services, from predictive analytics to proactive personalization.

5. **Quantum-Resistant Cryptography:** As quantum computing advances, there is a need for cryptographic methods that can withstand quantum attacks. This will be crucial for securing financial transactions in the future.
6. **Green Fintech Infrastructure:** This involves creating financial products and services that support environmental sustainability.
7. **Central Bank Digital Currencies (CBDCs):** Many central banks are exploring the development of digital currencies to enhance the efficiency and security of financial transactions.
8. **Autonomous Finance Platforms:** These platforms use AI and machine learning to automate financial decision-making and portfolio management, making professional-grade financial management accessible to everyday investors.

These emerging areas are set to redefine how financial services are delivered and experienced, making them more inclusive, efficient, and secure.

The adoption of fintech is being driven by several key factors:

1. **Technological Advancements:** Innovations in technology, such as blockchain, artificial intelligence, and machine learning, have enabled the development of new financial products and services that are more efficient and user-friendly.
2. **Consumer Demand:** Consumers are increasingly seeking convenient, fast, and secure financial solutions. The rise of smartphones and internet connectivity has made it easier for people to access financial services anytime, anywhere.
3. **Regulatory Support:** Many governments and regulatory bodies are creating supportive frameworks and policies that encourage the growth of fintech. This includes initiatives to promote financial inclusion and innovation.
4. **Cost Efficiency:** Fintech solutions often reduce operational costs for financial institutions by automating processes and reducing the need for physical infrastructure. This allows them to offer more competitive rates and services.
5. **Financial Inclusion:** Fintech is playing a crucial role in bringing financial services to unbanked and underbanked populations. Mobile banking and digital payment solutions are making it possible for people in remote and underserved areas to access financial services.
6. **Enhanced Security:** Advanced security measures and encryption technologies are building trust among consumers, making them more comfortable with using digital financial services.
7. **Personalization and Customer Experience:** Fintech companies are leveraging data analytics to offer personalized financial products and services that cater to individual needs and preferences, improving the overall customer experience.

8. **Competitive Pressure:** Traditional financial institutions are adopting fintech solutions to stay competitive in an increasingly digital landscape. They are partnering with or acquiring fintech startups to enhance their service offerings.

These drivers are collectively contributing to the rapid adoption and growth of fintech across the globe. While fintech brings many advantages, **there are also some disadvantages** and challenges that need to be considered:

1. **Security and Fraud Risks:** Despite advanced security measures, fintech platforms can still be vulnerable to cyberattacks, data breaches, and fraudulent activities. The digital nature of fintech makes it an attractive target for cybercriminals.
2. **Regulatory Uncertainty:** Fintech operates in a rapidly evolving regulatory environment. Compliance with varying regulations across different jurisdictions can be challenging, and sudden regulatory changes can impact business operations.
3. **Technical Issues:** Dependence on technology means that technical glitches, system outages, and software bugs can disrupt services and negatively affect user experiences.
4. **Limited Physical Interaction:** The digital-first approach of fintech reduces face-to-face interactions, which can be a disadvantage for customers who prefer in-person service and personalized assistance.
5. **Digital Divide:** Not everyone has access to smartphones, high-speed internet, or digital literacy skills. This digital divide can exclude certain segments of the population from benefiting fully from fintech services.
6. **Privacy Concerns:** The collection and analysis of large amounts of personal and financial data raise concerns about user privacy and data protection. Users may be wary of how their information is being used and stored.
7. **Market Saturation and Competition:** The fintech industry is highly competitive, with numerous startups and companies vying for market share. This can lead to market saturation and make it difficult for new entrants to succeed.
8. **Operational Risks:** Fintech companies, especially startups, may face operational risks due to limited resources, lack of experience, or reliance on third-party providers for critical services.

Understanding these disadvantages can help fintech companies and consumers navigate the challenges and make more informed decisions. Adopting fintech comes with several challenges that can hinder its widespread implementation. Here are some of the key challenges:

1. **Regulatory and Compliance Issues:** Navigating the complex and varied regulatory landscape across different regions can be daunting for fintech companies. Ensuring compliance with local laws, such as data protection and anti-money laundering (AML) regulations, adds to the complexity.

2. **Cybersecurity Concerns:** As fintech relies heavily on digital infrastructure, it is vulnerable to cyberattacks and data breaches. Ensuring robust cybersecurity measures is essential to protect sensitive financial information and maintain customer trust.
3. **User Trust and Adoption:** Gaining the trust of potential users can be challenging, especially for new fintech startups. Consumers may be hesitant to adopt new financial technologies due to concerns about security, privacy, and the reliability of the services.
4. **Infrastructure Limitations:** In some regions, inadequate internet connectivity and limited access to smartphones can hinder the adoption of fintech solutions. This is particularly true in rural and underserved areas.
5. **Integration with Traditional Systems:** Fintech solutions often need to integrate with existing legacy systems in traditional financial institutions. This can be a complex and costly process, requiring significant technical expertise and resources.
6. **Financial Literacy:** A lack of financial literacy among consumers can be a barrier to adopting fintech solutions. Educating users about the benefits and functionalities of these technologies is crucial for their widespread acceptance.
7. **Operational Risks:** Fintech companies, especially startups, may face operational risks such as limited funding, lack of experience, and challenges in scaling their operations. These factors can impact their ability to deliver consistent and reliable services.
8. **Competition:** The fintech landscape is highly competitive, with numerous players vying for market share. Standing out in a crowded market and attracting customers can be challenging for new entrants.

Despite these challenges, fintech continues to grow and evolve, driven by innovation and the increasing demand for digital financial services. Addressing these challenges through collaboration, regulatory support, and technological advancements will be key to the successful adoption of fintech.

Unit – 4

**Fintech – Banks &
Financial Institutions**

Traditional banks vs. Digital banks vs. Neo Banks

Based on the funding sources or types of customers, three basic bank business models can be defined:

- Retail-funded commercial bank
- wholesale-funded commercial bank
- capital-markets oriented bank.

Background: One crucial characteristic of the banking sector was the *lack of or the limited number of new entrants in the banking industry*. It was extremely difficult for new entrants to compete directly with highly developed and well-established banking institutions with a high level of capital and well-developed networks and infrastructure.

The favourable position of banks in the financial system was disturbed with the events induced by the financial crises that occurred in 2008, which revealed the major weaknesses in the way traditional banks conduct their operations. Consequently, four primary drivers brought changes in the business models of traditional banks, these are

- changes in customer expectations,
- Banking systems compatibility with the introduction of new financial products
- degree of profit volatility,
- increased competition and
- changes in the regulatory framework.

Changes in regulations and the simultaneous technological developments along with the other factors stimulated the birth of new type of competitors which challenged the traditional business models. Traditional brick and mortar bank business models with widespread networks of branches are becoming obsolete under the pressure of technological innovations where a new form of competitors emerged.

The digitization of financial services altered the existing business processes. It enabled the emergence of viable business models heavily relying on a different type of technology. Changes in the banking business model didn't result in change/modifications of the type of functions performed by banks; instead, the transformation was made on the way these functions are executed.

Traditional bank business model: Traditionally banks may operate under different business models and offer a variety of financial products and services where some banks may even be focused on becoming a one-stop-shop (bank) for its customers. Consequently, based on the customers they serve as well as their core products, traditional banks may be oriented toward retail clients or commercial clients or they may be operating under the universal bank model. Moreover, the primary funding source for traditional banks is their core deposits held by individual savers or by business entities. Furthermore, the main income source for

traditional banks is their interest income while they also generate non-interest income from their secondary activities and services. Another important characteristic of the traditional business model is that banks have a developed network of brick-and-mortar branches enabling a face-to-face interaction with their clients. Having a widespread branch network means that the traditional banks have high operating costs due to the large number of branches as well as costs associated with the need to maintain a well-established network of ATMs.

Digital bank business model: The digitalization of banking services has led towards the modification in the way banks communicate and serve their customers. Hence, the digital banking business model can be *considered as the crossover between the traditional and neobanks business model*. This model considers potential modifications in the structure of traditional banks with the application of innovative financial technology. Accordingly, the process of digital transformation considers the *modification of internal processes with the objective of their optimization*.

Neobanks business model: A neobank is a form of bank which is entirely online, and provides its products solely through online channels without owning physical branches. These banks offer their services through partnership with an incumbent bank or they may obtain a full banking license by going through the relevant application process. The majority of Neobanks customers are tech-savvy millennials who are confident in using new technology.

The essential bank business model that was established with the development of the *banking institutions was based on the physical delivery of its services through a network of branches*. However, the primary functions of the bank have not been changed, but the way these functions are executed and how products are delivered to clients has gone through some in-depth transformation.

The key objectives of transforming the bank business models are in terms of expanding the reach of distribution channels wide and deep in a cost efficient manner. Extensive reliance on innovative technologies have catalysed sales and distribution of Financial products and services.

The popularity of this model increased after the 2008 financial crises when financial products and services offered by traditional banks became available from institutions that didn't rely heavily on a branch network and this business model abandons the need for brick-and-mortar branches. Alternatively, although neobanks offer similar or the same financial products, this model has considerable differences when compared to traditional banks. The neobank model implements a low-cost structure offering feature-rich (and customized) products and services in an easy to access manner from any place throughout the day. Hence, it eliminates the need for clients to use the services or the need to visit the local branch during working hours. It is also stated that the model of Neobanks is developed to support three crucial areas which are *customer experience, user convenience and simplification of associated processes*.

Comparison of business models characteristics: It is evident that since the inception of the banking industry, the number of products offered by these financial institutions has been

broadening. However, during the last decade, the most prominent changes have occurred in terms of how bank products and services are delivered to the end customer and the type of institutions that deliver them. Accordingly, while any major difference in the financial services and products offered by the different banking business models cannot be firmly identified, there are some crucial differences in terms of the way in which they are brought closer to the end-user along with the depth of the product portfolio held under each of the three business models. The primary differences between the traditional business model and digital bank model are presented in table 2. The key differences may be viewed in terms of the period when the products are available, the time it takes to serve a customer, costs, geographical availability of services, and so on.

Table 2. Key differences between traditional and digital models of banking

Distinguishing features	Traditional model	Digital model
Customer service time frame	Limited. Service is carried out only at a clearly defined time	Unlimited. Possibility of round-the-clock access
The speed of customer service	Depends on the qualification and experience of the Bank employee	Immediate
Approach to service	Flexible, however, is limited to a small variety of service channels	Flexible and carried out through any convenient channel for the client
Maintenance cost	High, taking into account the bank's costs for the personnel and maintenance of departments	Low, often services are provided free of charge
Scope of service	Limited branching of the branch network and staffing	Unlimited, can go beyond the geographical location of the banking institution
Status of the operator in the service process	Functions of the operator is performed by an employee of the bank	Functions of the operator are performed by the bank's client
The procedure for learning new services and promotions	Requires time and cost	Carried out quickly, via SMS and e-mail newsletter
Consumable component of the operation of the service system	The key models are articles on the staff and maintenance of departments	The key articles are articles for the purchase and maintenance of servers and software package

Differences between Traditional model and Neo model in Banking

Incumbents' Model Product-centric	Neo-banks' Model Platform-centric
Branches	Marketplace community
High verticalization of products	High community engagement with the product
No strong communities around the brand/product	Horizontalization of products with best-in-class apps
Large branch/network	Open banking through APIS
High customer acquisition costs	Mobile only (low customer acquisition cost)
Tech infrastructure	Tech infrastructure
Outsourced technology	Open architecture technology stack
Rigid legacy platforms of 1970s–1980s	Ability to 'plug and play' best-in-class tools/services
Non-modular architecture with limited scalability	Low dependency on 3rd party providers
Hard to integrate data from 3rd party providers	Open ecosystem allowing access to external developers
PRODUCT-CENTRIC BUSINESS	DATA-DRIVEN BUSINESS
Banking data divided into silos	Real-time data integration and management
Mix of paper-based and online data	Current accounts generate key data for 3rd parties
Difficulties in reconciling, monitoring data	Data powers business model through 3rd party affiliation fees
ORGANIZATIONAL DESIGN	ORGANIZATIONAL DESIGN
Product-centric organizational design	Customer-centric organizational design
Products and verticals drive organization structure	Customer needs drive data architecture

Criteria	Neobanks	Digital Bank	Conventional Bank
Regulated by a Bank?	Not regulated by banks (they follow RBI and govt. regulation)	Regulated	Regulated
Customer Support	Online	Online	In-person, online
Banking License	None/partial/full	Partial	Full
Service Platforms	Web services and mobile apps	Web services and mobile apps	Physical establishment
Application Process	Online	Online	In-person
Approval Process	Quick	Quick	Manual and time-consuming

List of Neobanks in India

1. Akudo	2. Chqbook	3. Digibank	4. FamPay	5. FI Money
6. Finin	7. Freo	8. InstantPay	9. Jupiter	10. Mahila Money
11. Mool	12. Niyo	13. OneCard	14. Open	15. Paytm Bank
16. Piggy Neobank		17. RazorpayX	18. Uni Card	19. Walrus Club
20. ZikZuk	21. Zolve			

Conclusion

Neobanking has brought fresh air to the way traditional banking works in India. It's giving birth to new [innovative ideas](#) to save, manage and intensify money in the upcoming time. Various businesses and individuals benefit from such banking activities. There lie multiple reasons behind its overall success rate in India. One of the primary reasons is the personalized & customized financial services the neo banks provide. Another reason is the handy, convenient, and latest technology adoption. Neobanks in India have proved to be a big-time game changer, whereas traditional banks are trying their best to compete in this race.

With its excellent savings, tailored financial solutions, no-cost banking services, operational ease, and latest technological advancements, this list of Neobanks in India has gained immense popularity. Neo banks will undoubtedly take over all traditional ways of handling finances. This phenomenon has grown massively in India in the last few years. Also, while counting the list of industry leaders, neo banks would occupy the number 1 rank in the upcoming few years.

Technology adoption by Banks: Andhra Bank introduced the first credit cards in 1981. This was 30 years after the first bank credit card was introduced in New York. Soon, many other banks in India followed suit and issued their own credit cards. In 1987, HSBC Bank set up the first ATM. BillDesk was founded in 2000, on the cusp of the new decade. It was the first payment aggregator in the country, and it made digital payments easier for e-commerce customers.

Key challenges faced by banks: Inflation, fluctuating interest rates, global economic uncertainty, and regulatory reforms; banks are under immense pressure to remain resilient and profitable while adhering to complex compliance requirements – including anti-money laundering (AML), know your customer (KYC) and the General Data Protection Regulation (GDPR).

Augmented by those regulatory reforms and wave of enforcement action coupled with increasing competition from non-banks, combined, they influence everything from cash flow management and credit availability, to more precise credit assessments; all leading to slower processing times, increased rejection rates, and reduced risk appetites – potentially stifling economic growth further. Technological advancements are essential to maintaining a

competitive advantage, especially in more traditional environments that have typically lagged behind the pace of change due to organizational complexities, or those who are hesitant about the regulatory safety of AI tools.

Evolving technologies like AI, SaaS (Software as a service), machine learning, and APIs are transformative in their ability to help banks manage risks more effectively, remove silos, streamline operations, and enhance decision-making processes. Using these technologies, offers an ability to collate multiple data sources, identify patterns, and manage organization's risks more holistically, uncovering opportunities for growth while automating routine tasks. By connecting various risk factors, banks can proactively mitigate these risks and bring efficiencies that elevate decision-making processes by freeing up human resources to consider and validate the outputs and think more strategically.

Impact of technology in Banking:

- Enhancing credit risk analysis
- AI and machine learning as an investment strategy
- Tackling fraud & cybersecurity threats
- Future outlooks for successful technology integration
- Decision confidence
- Targeted use of expertise
- Differentiated customer experience
- Understanding younger, tech-savvy borrowers
- Delivering faster, frictionless experiences

The disruptive impact of AI

Banks need to separate risk signals from the noise, connect data more seamlessly, and uncover patterns hidden within the chaos so they can stay ahead of the curve.

Chatbots: A chatbot is an artificial intelligence (AI) program designed to simulate conversation with users, typically via text or voice interactions.

Components of a Chatbot

- **User Interface:** This is the platform where users interact with the chatbot, such as a website chat window, a messaging app, or a voice assistant.
- **Natural Language Processing (NLP):** Chatbots use NLP to understand and process the user's input. NLP enables the bot to understand human language, interpret the meaning, and respond appropriately.
- **Knowledge Base:** This is a repository/library of information that the chatbot uses to provide answers. It could include predefined responses, databases, or links to external resources.

- **Machine Learning:** Many chatbots use machine learning algorithms to improve their responses over time. By analyzing past interactions, the bot can learn and adapt to provide more accurate and relevant answers.
- **Context Management:** This allows the chatbot to remember and reference previous interactions to maintain context throughout the conversation. This makes the interaction more coherent and personalized.

Types of Chatbots

- **Rule-Based Chatbots:** These follow predefined scripts and rules. They can handle straightforward queries but may struggle with complex or nuanced conversations.
- **AI-Powered Chatbots:** These use advanced AI, including machine learning and NLP, to understand and respond to a wide range of queries. They can handle more complex interactions and learn from past conversations. Summary:

Aspect	Rule-Based Chatbots	AI-Powered Chatbots
Structure	Predefined rules and scripts	Machine learning and NLP
Functionality	Limited scope, static responses	Adaptive responses, learning capability
Implementation	Easier setup, cost-effective	Advanced setup, higher cost
Use Cases	FAQs, simple transactions	Customer support, e-commerce assistance, complex queries

Applications of Chatbots

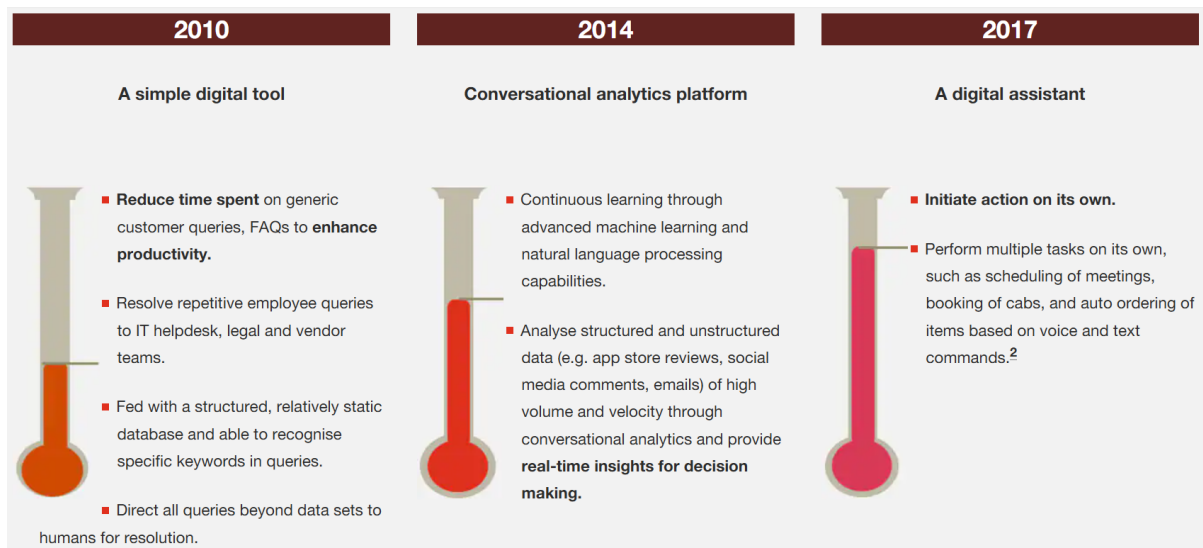
- **Customer Service:** Chatbots can provide instant support to customers by answering queries, guiding them through processes, or even resolving issues.
- **E-commerce:** Bots can assist shoppers by recommending products, processing orders, or providing updates on deliveries.
- **Healthcare:** Chatbots can offer medical advice, remind patients about appointments or medications, and help with symptom checking.
- **Entertainment:** Some chatbots are designed for fun and engagement, offering games, trivia, or casual conversation.
- **Personal Assistants:** These can help manage schedules, set reminders, and perform tasks like sending messages or making calls.

Benefits of Chatbots

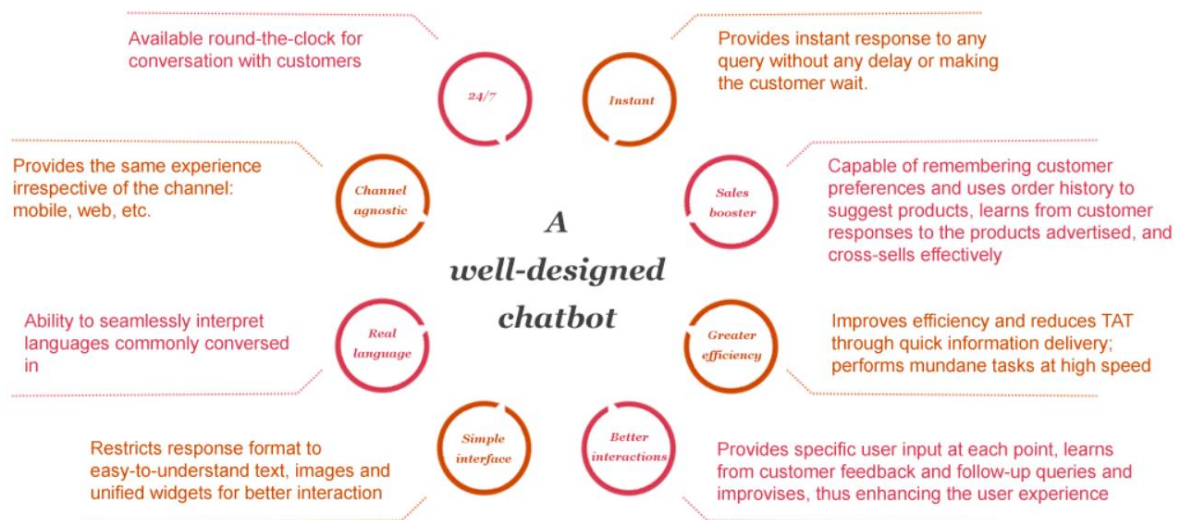
- **24/7 Availability:** Chatbots can provide support round the clock without the need for human intervention.
- **Efficiency:** They can handle multiple interactions simultaneously, reducing wait times for users.
- **Cost-Effective:** Deploying chatbots can reduce the need for a large customer service team, cutting operational costs.
- **Consistency:** Bots provide consistent responses based on their programming and knowledge base, ensuring uniformity in information dissemination.

Chatbots have become increasingly sophisticated, and their applications are expanding across various industries. Whether you're chatting with a customer service bot or playing a trivia game, chatbots are changing the way we interact with technology.

Chatbots – Evolution: Raising the bar of intelligence over time



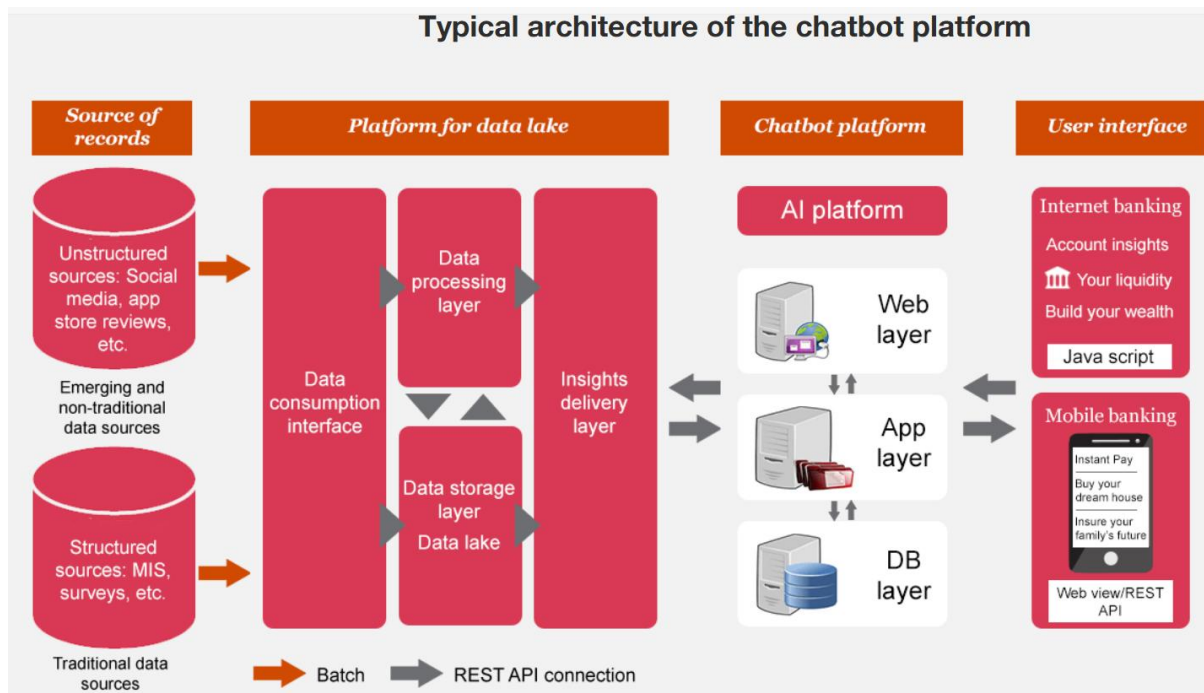
Characteristics of a well-designed chatbot



Chatbots are employed for the following functions:

- Automate basic banking activities
- Answer FAQs
- Troubleshoot issues
- Send reminders
- Recommend products
- Provide insights and analytics

Increasingly, chatbots are also attracting interest in the world of FinTech, and a number of companies have developed their own chatbots using **proprietary technology and algorithms**. Chatbots utilise application programming interfaces (APIs) to integrate with data management platforms. This allows them to analyse the extracted data as well as web- and mobile-based user interfaces and deliver the necessary insights to the end customer.



Adoption of chatbots by financial institutions

- Proactive suggestions for superior customer satisfaction
- Speedy authentication
- CXO dashboard

Payment Systems:

A payment system is essentially a framework that facilitates the transfer of money. It allows financial transactions to occur between individuals, businesses, and institutions.

There are several key components to a payment system:

1. **Participants:** These are the entities involved in the transaction, such as buyers, sellers, banks, and payment service providers.
2. **Instruments:** These are the tools used to initiate and complete transactions, such as cash, checks, credit/debit cards, and digital wallets.
3. **Processing:** This involves the steps taken to authorize, clear, and settle transactions. Authorization ensures that the payer has enough funds, clearing confirms the details of the transaction, and settlement is the actual transfer of funds.
4. **Regulations and Standards:** Payment systems operate under a set of rules and standards to ensure security, efficiency, and interoperability. These regulations can be set by governmental bodies, international organizations, or industry groups.

5. **Infrastructure:** This includes the physical and digital networks that support the transfer of funds, such as banking networks, payment gateways, and cybersecurity measures.

Payment systems have evolved significantly, especially with the advent of digital technology. Today, we have various types of payment systems including:

- **Traditional Banking Systems:** Transactions conducted via bank branches or ATMs.
- **Card Payment Systems:** Transactions involving credit, debit, or prepaid cards.
- **Online Payment Systems:** Transactions conducted through the internet using digital wallets, payment gateways, or online banking.
- **Mobile Payment Systems:** Transactions conducted using mobile devices through apps or contactless payments.
- **Cryptocurrency Payment Systems:** Transactions conducted using digital currencies like Bitcoin.

These systems play a crucial role in facilitating commerce and enabling economic activities. They provide the necessary infrastructure for the smooth operation of the financial system, promoting trust and reliability among participants.

Digital Payment Systems:

Digital payment systems have revolutionized how we handle transactions. Here's a step-by-step breakdown of how they work:

1. **Initiation:** The process begins when a payer initiates a transaction. This could be through a mobile app, a website, or a physical point-of-sale terminal. For example, you might scan a QR code at a café or enter your card details online while shopping.
2. **Authorization:** The payment request is sent to the payer's bank or payment service provider. The system checks if the payer has sufficient funds or credit available. If the payer uses a credit or debit card, this might involve entering a PIN or a one-time password (OTP) for verification.
3. **Encryption:** To ensure security, the transaction details are encrypted. Encryption converts the information into a secure code, which can only be deciphered by the intended recipient. This step prevents fraud and unauthorized access.
4. **Transmission:** The encrypted transaction details are transmitted to the payment processor, which acts as an intermediary between the payer's and the payee's banks. The processor verifies the details and forwards the request to the payer's bank.
5. **Clearing and Settlement:** The payer's bank clears the transaction by debiting the payer's account and crediting the payee's account. Settlement ensures that the funds are transferred from one account to another. This process might take a few seconds to a couple of days, depending on the payment method.

6. **Notification:** Both the payer and the payee receive notifications confirming the transaction. This could be an SMS, email, or in-app notification. The payee can now provide the goods or services to the payer.

Key Technologies Involved

Technology adoption in payment systems has brought about significant changes and innovations in how we handle transactions. Here's an overview of some key developments:

1. **Mobile Payments:** With the proliferation of smartphones, mobile payment systems like Apple Pay, Google Wallet, and Samsung Pay have gained popularity. These systems allow users to make payments using their mobile devices, often through contactless technology.
2. **Digital Wallets:** Digital wallets store payment information securely and can be used for online and in-store purchases. Popular digital wallets include PayPal, Venmo, and Cash App.
3. **Blockchain and Cryptocurrencies:** Blockchain technology has introduced a new level of security and transparency to payment systems. Cryptocurrencies like Bitcoin and Ethereum offer decentralized payment options that can bypass traditional banking systems.
4. **Contactless Payments:** Contactless payment methods, such as NFC (Near Field Communication) and RFID (Radio Frequency Identification), enable users to make payments by simply tapping their card or device on a payment terminal. This technology is widely used in credit and debit cards, as well as in mobile payment apps.
5. **QR Code Payments:** QR code payments have become popular in many countries, especially in Asia. Users can scan a QR code to make payments quickly and easily, often through mobile payment apps.
6. **Buy Now, Pay Later (BNPL):** BNPL services, such as Klarna and Afterpay, allow consumers to make purchases and pay for them in installments. This has become a popular payment option for online shopping.
7. **Peer-to-Peer Payments:** Peer-to-peer (P2P) payment systems, like Venmo and Zelle, enable users to transfer money directly to one another without the need for cash or checks. These systems are often integrated into mobile apps and online banking platforms.
8. **Biometric Payments:** Biometric authentication, such as fingerprint or facial recognition, is increasingly being used to secure payment transactions. This adds an extra layer of security and convenience for users.

The adoption of these technologies has not only increased the convenience and efficiency of payment systems but also opened up new opportunities for businesses and consumers alike.

Advantages & contribution of Digital Payment Systems

- **Convenience:** Transactions can be made anytime and anywhere with an internet connection.
- **Speed:** Digital payments are typically faster than traditional methods like cash or checks.
- **Security:** Advanced encryption and authentication protocols reduce the risk of fraud.
- **Traceability:** Digital payments leave a clear transaction trail, which can be useful for budgeting and record-keeping.
- **Regulatory Changes:** Governments and regulatory bodies are adapting to these technological advancements by updating regulations to ensure consumer protection and promote fair competition.
- **Financial Inclusion:** Technology adoption in payment systems has helped to increase financial inclusion by providing access to banking services for unbanked and underbanked populations. Mobile banking and digital wallets are particularly effective in reaching these groups.

Digital payment systems continue to evolve, offering increasingly seamless, secure, and innovative ways to transact.

Settlement - Concept

In the banking context, settlement refers to the process of transferring funds or securities to complete a financial transaction. Here are some key aspects of settlement in banking:

1. **Interbank Settlement:** This involves the transfer of funds between different banks. Systems like RTGS (Real-Time Gross Settlement) and NEFT (National Electronic Funds Transfer) are used for this purpose. RTGS is used for large-value transactions that require immediate settlement, while NEFT is used for smaller transactions and settles in hourly batches.
2. **Securities Settlement:** In stock exchanges, settlement refers to the process of transferring securities and funds between buyers and sellers to complete trades. Clearing corporations or clearinghouses act as intermediaries to ensure the timely and efficient settlement of trades.
3. **Merchant Settlement:** This involves the transfer of funds from the cardholder's bank (issuing bank) to the merchant's bank (acquiring bank) for transactions processed through debit cards, credit cards, or other electronic payment methods.
4. **Loan Settlement:** This involves the repayment of principal and interest amounts by borrowers to lenders according to the terms of the loan agreement. In cases of loan settlements or restructuring, banks may negotiate with borrowers to settle outstanding debts or modify the terms of the loan.

5. **Foreign Exchange Settlement:** This involves the exchange of currencies between parties to complete international trade or investment transactions. Systems like CLS (Continuous Linked Settlement) are used to mitigate settlement risk and ensure the timely completion of foreign exchange transactions.

Settlement is a crucial process in the banking system as it ensures that transactions are completed accurately and efficiently, reducing the risk of default and maintaining the stability of the financial system. Several technologies facilitate settlement processes in banking, enhancing efficiency, security, and speed. Here are some **key technologies**:

1. **Blockchain Technology:** Blockchain provides a decentralized and tamper-proof ledger for transactions. It enables real-time settlement, reduces the risk of fraud, and enhances transparency. For example, JPMorgan and several Indian banks are testing a blockchain-based settlement system to expedite dollar transactions.
2. **Real-Time Gross Settlement (RTGS):** RTGS systems allow for the immediate transfer of funds between banks on a real-time and gross basis. This means transactions are settled individually and instantly, reducing settlement risk.
3. **Automated Clearing House (ACH):** ACH networks facilitate batch processing of transactions, allowing for the electronic transfer of funds between banks. This system is commonly used for payroll, direct deposits, and bill payments.
4. **Continuous Linked Settlement (CLS):** CLS is a global multi-currency settlement system that mitigates settlement risk in foreign exchange transactions. It ensures that both sides of a transaction are settled simultaneously.
5. **Near Field Communication (NFC):** NFC technology enables contactless payments by allowing devices to communicate when they are in close proximity. This technology is widely used in mobile payment systems and contactless cards.
6. **Artificial Intelligence (AI) and Machine Learning (ML):** AI and ML are used to detect and prevent fraudulent transactions, optimize settlement processes, and enhance decision-making in real-time.
7. **Application Programming Interfaces (APIs):** APIs enable seamless integration between different banking systems and third-party applications, facilitating faster and more efficient settlement processes.
8. **Distributed Ledger Technology (DLT):** Similar to blockchain, DLT provides a decentralized database that is shared across multiple sites, enhancing the security and transparency of settlement processes.

These technologies are transforming the banking industry by making settlement processes more efficient, secure, and reliable.

Regtech

RegTech, short for Regulatory Technology, is a rapidly evolving ecosystem designed to help businesses, especially in highly regulated industries like finance to comply with regulations more efficiently and cost-effectively. Here's a detailed breakdown of its ecosystem

Core Components of the Regtech EcoSystem

Regulatory Reporting: Tools that automate the collection, analysis, and submission of data to regulatory authorities. This reduces manual errors and ensures timely compliance.

Risk Management: Solutions that identify, assess, and mitigate risks, such as financial fraud, cybersecurity threats, and operational risks.

Identity Management and Verification: Technologies like biometrics, blockchain, and AI are used to verify identities, prevent fraud, and ensure secure transactions.

Transaction Monitoring: Real-time monitoring systems that flag suspicious activities, such as money laundering or unauthorized transactions.

Compliance Support: Platforms that provide guidance on regulatory requirements and help businesses adapt to changing laws.

Technologies Driving RegTech

Artificial Intelligence (AI): Used for predictive analytics, fraud detection, and automating compliance tasks

Blockchain: Ensures secure and transparent record-keeping, especially for identity verification and transaction tracking.

Cloud Computing: Facilitates scalable and cost-effective data storage and sharing.

Big Data Analytics: Helps analyze vast amounts of data to identify patterns, risks, and compliance gaps.

Key Stakeholders

RegTech Companies: Innovators providing solutions to address regulatory challenges.

Financial Institutions: Major users of RegTech to ensure compliance with financial regulations.

Regulators: Collaborate with RegTech firms to create a more transparent and efficient regulatory environment.

Benefits of RegTech

Cost Efficiency: Reduces the need for extensive manual processes and consultancy fees.

Improved Accuracy: Minimizes errors in compliance reporting and risk assessment.

Real-Time Monitoring: Enables businesses to detect and respond to issues promptly.

Adaptability: Helps organizations stay ahead of evolving regulations.

Challenges in the RegTech Ecosystem

Integration with Legacy Systems: Many organizations struggle to integrate new technologies with their existing infrastructure.

Data Privacy Concerns: Handling sensitive data requires robust security measures.

Regulatory Uncertainty: Frequent changes in regulations can make it challenging for businesses to keep up.

The RegTech ecosystem is continuously expanding, driven by advancements in technology and the increasing complexity of regulatory requirements. It plays a crucial role in fostering trust, transparency, and efficiency in the modern business landscape.

Unit 5 – Digital Currencies

Digital Currencies

Cryptocurrencies are digital assets that rely on an encrypted network to execute, verify, and record transactions, independent of a centralized authority such as a government or bank.

Key Points

- Cryptocurrency was developed as an alternative to the dollar, and its functions can make it an attractive investment.
- Blockchain, the underlying technology that powers crypto, is seen as a tech disruptor.
- Much like dot-com investing in the 1990s, crypto holds promise, but there would be winners and losers.

Characteristics:

- Cryptocurrencies (or “crypto” for short) are **decentralized currencies**, meaning they’re neither issued nor governed by a central bank. Some cryptocurrencies are issued by their developers, while others are generated by their respective network algorithms.
- Crypto are **digital assets**—they have no tangible form.
- Cryptocurrencies exist and **operate on a public ledger** called a blockchain, which records all crypto transactions.
- Blockchain encryption is designed to make all transactions **immutable and secure** from tampering, counterfeit, and other forms of fraudulent transactions.

Although cryptocurrency is defined as a form of “digital currency”—implying it’s a kind of money—most businesses and consumers have not adopted it as a common medium of exchange. In other words, most stores will not accept crypto as a form of payment.

[Bitcoin](#) may be an exception, as some businesses have accepted it as payment for goods and services. So, if crypto isn’t a common form of money, why do people buy it?

- **It’s an alternate asset class.** Although some crypto investors are hopeful that cryptocurrencies might someday be adopted as a form of money, most see crypto as an alternative asset that can appreciate in value.
- **It’s a way to invest in blockchain technology.** Some people purchase cryptocurrency as a way of indirectly investing in its underlying blockchain.

Types of Digital currencies:

- **Crypto Currency:** Crypto currencies are digital currencies that use cryptography to secure and verify transactions in a network. Cryptography is also used to manage and control creation of cryptocurrencies. Normally cryptos are not regulated.
- **Central Bank Digital currency (CBDC):** As the name suggests, CBDCs are regulated digital currencies issued by the Central Bank of a country. CBDCs supplement the fiat/physical currencies in a country and are scaled up to gradually replace physical currencies.

- **Virtual Currencies:** Unregulated digital currencies extensively used on gaming platforms, and controlled by developers or founding organizations.

Pros of DCs

- Faster transaction times
- No physical manufacturing required
- Lower transaction costs
- Make it easier to implement monetary and fiscal policy
- Greater privacy than other forms of currency

Cons of DCs

- Can be difficult to store and use
- Prone to hacking
- Volatile prices that result in lost value
- May not allow for irrevocability of transactions
- Limited acceptability

Differentiation between Crypto Currency and Digital Currency

Crypto Currency

- Not regulated by any Regulatory authority
- No control over the supply, though there may be overall cap imposed to mine
- Can be used for investment or speculation
- The value of cryptos are prone to volatility and extreme price swings
- All cryptos are digital currencies. Not necessary all DCs are cryptos.
- Traded on exchanges

Central Bank Digital Currency

- Regulated by Central Bank of a country
- The central bank has complete control over the issue and supply of CBDC.
- Normally used for p2p payments and settlement of commercial transactions.
- The value of currency is limited to the face value of the currency
- Not traded in exchanges. Used as a means of settling dues arising out of transactions

Most popular types of cryptocurrencies?

Among the 18,000-plus cryptocurrencies in existence, **Bitcoin** and **Ethereum** are the two largest cryptocurrencies by [market capitalization](#). Bitcoin, the original and largest cryptocurrency, was developed in 2009 as an alternative monetary asset. It was meant to be an alternative to the U.S. dollar and other [fiat currencies](#). Although some vendors may accept Bitcoin as payment, most investors view it as a speculative investment.

Ethereum is the second-largest cryptocurrency by market cap. Unlike Bitcoin, Ethereum was not designed to function solely as an alternative monetary asset. Instead, it was designed as an innovative ledger technology to help companies securely transport data, store data and build new programmes and applications.

In short, Ethereum is a massive digital ecosystem through which digital information and computer applications can be transported, stored, and even created.

What does crypto aim to achieve?

Some cryptocurrencies, like Bitcoin and **Tether**, were developed to serve a monetary function. Others, such as **Dogecoin** and **Shiba Inu** coin, were developed as novelty items whose values rely on popularity and trading.

Many, if not most, cryptocurrencies were developed to solve challenges within the blockchain ecosystem, such as transmission speed, scalability, security, energy efficiency and cost efficiency.

What are the risks of investing in crypto?

Here are a few of the more common risks in cryptocurrency investment:

- **Volatility risk.** Crypto prices frequently exhibit extreme swings during certain economic or market conditions.
- **Liquidity risk.** Some cryptocurrencies trade with light volume, and thus can be easily manipulated by buyers with large capital resources or Sellers who have a large stake in a given currency.
- **Cybersecurity risk.** Crypto can be stolen if a bad actor has access to crypto wallet's private key.
- **Overnight risk.** Because crypto trades 24/7, holdings are subject to adverse fluctuations overnight.
- **Vanishing risk.** There are factors that have caused certain crypto coins to vanish; these instances are rare and unique to particular coins.

Crypto currency exchanges in India

- Zebpay
- Unocoin
- Coinsecure

- Bitcoin-India
- Btcxindia
- Ethexindia
- Bitxoxo
- Gatehub
- Binance India
- WazirX India
- Coinbase India
- CoinDCX
- Kraken India

Cryptocurrency is a form of digital or virtual currency that relies on cryptography for security. Unlike traditional currencies issued by governments (fiat currencies), cryptocurrencies operate on decentralized networks based on blockchain technology—a distributed ledger enforced by a network of computers (nodes). Bitcoin, launched in 2009 by an anonymous entity known as Satoshi Nakamoto, was the first cryptocurrency and remains the most well-known and widely used.

Cryptocurrencies are designed to function as a medium of exchange, but they also serve other purposes, such as a store of value and a means of transferring assets across borders. The decentralized nature of cryptocurrencies makes them resistant to government interference or manipulation, a characteristic that has drawn both supporters and critics.

How Does Cryptocurrency Work?

Cryptocurrencies work on a technology called blockchain. A blockchain is a decentralized network of computers that work together to record and verify transactions. Here's a basic overview of how it works:

1. Blockchain: A blockchain is a public ledger that records all cryptocurrency transactions. Each transaction is grouped into a block, and once the block is complete, it is added to the chain of previous blocks, hence the name "blockchain." The blockchain is maintained by a network of nodes (computers) that ensure its integrity.

2. Decentralization: Unlike traditional banking systems that are centralized, where a single entity has control, cryptocurrencies operate on decentralized networks. No single entity controls the blockchain, making it more secure and transparent.

3. Mining: Mining is the process through which new cryptocurrency coins or tokens are created. It involves solving complex mathematical problems, and miners are rewarded with cryptocurrency for their efforts. Bitcoin, for example, has a fixed supply of 21 million coins, and mining becomes progressively more difficult as more coins are mined.

4. Private and Public Keys: Every cryptocurrency wallet has a pair of keys—a public key (like an account number) and a private key (like a password). The public key is used to receive funds, while the private key is used to sign transactions and access the wallet.

5. Transactions: When you send cryptocurrency to someone, the transaction is broadcast to the network. The miners verify the transaction, ensuring the sender has sufficient funds and that the transaction is legitimate. Once verified, the transaction is added to the blockchain, making it immutable and transparent.

Characteristics of Blockchain Technology

1. Decentralization
2. Immutability
3. Transparency
4. Security
5. Consensus mechanism
6. Smart contracts
7. Distributed Ledger
8. Traceability
9. Tokenization

How to Buy Bitcoin or Cryptocurrency in India?

Buying cryptocurrency in India is relatively straightforward. You can purchase Bitcoin or other cryptocurrencies through various exchanges. Here are some popular cryptocurrency exchanges where you can buy, sell, and trade cryptocurrencies in India:

WazirX: One of the most popular cryptocurrency exchanges in India, WazirX allows users to buy and trade a wide range of cryptocurrencies. It also offers a P2P trading platform.

CoinDCX: Another leading Indian exchange, CoinDCX offers a user-friendly interface, a variety of cryptocurrencies, and the option to trade with minimal fees.

ZebPay: An older and well-established exchange in India, ZebPay allows you to buy Bitcoin and other cryptocurrencies. It also offers a mobile app for trading on the go.

CoinSwitch Kuber: A simple and beginner-friendly platform, CoinSwitch Kuber allows users to buy over 100 cryptocurrencies with INR and has an easy-to-use interface.

UnoCoin: UnoCoin is one of the oldest Bitcoin exchanges in India, offering a variety of services, including buying, selling, and merchant gateways for businesses that accept Bitcoin.

